Rameen Assigned Source Review

Operating Systems CEP | Topic 23 | S015-S020 | 22 October 2026

Source review prepared at Raazia's request for Rameen's assignment. This package is a first source check and audit, not Rameen's independent human signoff. Original master assessments are preserved; score changes below are proposals for adjudication.

Review item	Checked	Pending
Full-text sources	6	0
Eligibility criteria	36	0
QA items	60	0
Extracted outcome rows	9	0

All six assigned reports have source checks. The correct InferFair publisher copy closes the final missing source. Corrections are integrated into the master. The559retained-record selection branch and independent human signoff remain open.

Key findings

RM01 | P015 | section8 case3; PDFp14/printedp683 Evidence: Printed source heading

RM02 | P015 Q9 | Proposed2/2; compute/performance control is claimed and tested Evidence: PDFpp14-15; Figure15a; section9. Submitted Q9 is not restricted to security/fault guarantees.

RM03 | P021 Q9 | Proposed2/2 for delimited compute-resource boundary Evidence: PDFpp10-12; KRISP-I/O; Figures13,16. No stronger security claim inferred.

RM04 | P021 Q10 | Proposed2/2 for transparent replication and limitation details Evidence: PDFpp7-9; emulation calibration and formula. Emulation alone is not a failure of reporting.

RM05 | P022 | Pin arXiv2410.07381v3; 17pages; DOI-matched ASPLOS manuscript Evidence: v3PDFp1 and arXiv version history

RM06 | P022 E052 | Normalized total system throughput Evidence: v3PDFpp9-10; sections5.1,5.3

RM07 | P022 Q5 | Proposed2/2; stated policies and common conditions support interpretable comparison Evidence: v3PDFpp9-11. Different policy objectives alone do not invalidate a benchmark comparison.

RM08 | P022 Q9 | Proposed2/2 for directly measured performance isolation Evidence: v3PDFpp9-13; no security/fault equivalence inferred

RM09 | P018 P019 P020 | Historical access gap closed:allthreepublisher PDFs now supplied andchecked;original unavailable status retained inhistory. Evidence: uploaded_source_receipt.json;inferfair_receipt.json

RM10 | Novelty | Position as bounded, source-audited, condition-specific update; no first-review claim Evidence: Yu2022; Huang2025; Ye2024; supplementary search receipt

RM11 | Recommendations REC04-REC06 | REC04-REC06 primary-source checks completed;retain experiment-specificconditions andlimits. Evidence: Scope/recommendation audit

RM12 | P022 | ONNX Runtime GPU1.17.0 Evidence: v3PDFp9; section5.1

RM13 | P019 P020 | Historical22October initial checkpoint: ArkGPU publisher PDF supplied; the then-pending InferFair gap was later closed by RM16 and RM17. Evidence: PDF title pages; uploaded_source_receipt.json

RM14 | P020 E050 | Section4.2; PDFpp13-14; Figure12 onp16 Evidence: Exact surrounding heading and paragraph verified

RM15 | P019 E048 E049 | P99 pair count: PDFp11; QPS loss: PDFp12 Evidence: Section5.2 and Figures14/16

RM16 | P018 | Full11page article inspected;E046/E047 values andlocalcontext confirmed Evidence: PDFp9;Table3

RM17 | P018 QA | 17/20current source appraisal:Q4lowered for missingstackversions;Q5/Q7/Q9supported byexplicitcomparators,SLOmethod andfairnessboundary Evidence: QA_Source_Review.csv;submittedrubric

S015 / P015

XSched: Preemptive Scheduling for Diverse XPUs

Access status: Full text inspected

Version checked: USENIX proceedings PDF with cover; printed pp671-692

DOI: USENIX proceedings (no DOI assigned)

Source URL: https://www.usenix.org/system/files/osdi25-shen-weihang.pdf

Original QA: 16/18. Proposed QA: 18/20.

Criterion	Review finding and source
IC1	Supported in accessed version. English final USENIX proceedings identity verified. PDF cover and pp2,23
IC2	Supported in accessed version. Temporal/API scheduling among concurrent GPU inference clients. PDF pp4-10; sections3-6
IC3	Supported in accessed version. Two BERT-large inference clients; non-GPU cases excluded from extraction. PDF p14; section8 case3
IC4	Supported in accessed version. P99 ratio and throughput-latency evidence inspectable. PDF p14; Figure15a
IC5	Supported in accessed version. Single-GV100 inference case is separately reported. PDF p14; single GV100 experiment
IC6	Supported in accessed version. 2025 lies within the main publication window. PDF cover: OSDI2025

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Portable preemptive scheduling objective and GPU-serving case are explicit. PDF pp2-3; section1
Q2	2	2	XQueue, XShim and three preemption levels specify the mechanism. PDF pp4-10; sections3-6
Q3	1	1	Setup and policies are described; repeated-run uncertainty for Figure15 is not specified. PDF pp11-14; sections7-8
Q4	1	1	GV100, BERT-large and client rates are explicit; exact Triton/CUDA versions and batch sizes for case3 remain unresolved. PDF p14; section8 case3
Q5	2	2	Same-GPU standalone, native and priority Triton cases use the stated client conditions. PDF p14; Figure15a
Q6	2	2	Throughput in requests/s and normalized P99 comparisons are distinguished. PDF p14; Figure15
Q7	2	2	P99, latency CDF, high-priority 10 requests/s and continuous background load are explicit. PDF p14; section8 case3; Figure15a
Q8	2	2	Two BERT-large clients share one GV100 in the retained experiment. PDF p14; section8 case3
Q9	NA	2	Direct foreground performance-control evidence is delimited; malicious bypass is explicitly a limitation, not a security guarantee. PDF pp14-15; Figure15a; section9 Untrusted tenants
Q10	2	2	Artifact instructions, hardware-dependent preemption limits and sufficient-memory assumption are documented. PDF pp15,23; section9; Artifact Appendix

Outcome checks

E044 | Reported high-priorityP99 latency ratio: 1.07 x standalone. Value and context checked.

Baseline: standalone high-priorityBert-large execution. Hardware: single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes. Workload: twoBert-large clients ononeGV100;high-priority10req/s,backgroundcontinuous.

Triton+XSched;not a rawmillisecond value. NativeTriton ratio1.53x;noNPUresultincluded. 1.07x standalone and author-rounded 30.0% decrease are textual results; section7.4 label in inherited anchor corrected. Source anchor: PDF p14; printed p683; section8 case3; Figure15a

E045 | Author-reported P99 latency reduction: 30.0 percent. Value and context checked.

Baseline: vanillaTriton (P99=1.53xstandalone). Hardware: single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes. Workload: same two-clientTriton/GV100case.

Within-paper comparison only;author-rounded30.0%,not a universal bound. 1.07x standalone and author-rounded 30.0% decrease are textual results; section7.4 label in inherited anchor corrected. Source anchor: PDF p14; printed p683; section8 case3; Figure15a

S016 / P018

InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

Access status: Full text inspected

Version checked: User-supplied Elsevier typeset article;FGCS150(2024),10-20;DOI matched onPDFp1

DOI: 10.1016/j.future.2023.08.020

Source URL: https://www.sciencedirect.com/science/article/pii/S0167739X23003187

Original QA: 15/20. Proposed QA: 17/20.

Criterion	Review finding and source
IC1	Supported in accessed version. English journal primary study;typeset DOI/title/year match. PDFp1
IC2	Supported in accessed version. TemporalSLO-aware scheduling with feedback goodput balancing. PDFpp5-8;section4
IC3	Supported in accessed version. Concurrent heterogeneous DNN inference;three-model Table3 workload. PDFpp2,9;Tables1,3
IC4	Supported in accessed version. InspectableSLO-compliant goodput and fairness with definitions. PDFpp2,8-9;Eq1-2;Table3
IC5	Supported in accessed version. Local co-location tests separately reported fromfourGPU realistictrace. PDFp9;sections5.3-5.4
IC6	Supported in accessed version. 2024issue year within2020-2026window;online date18August2023 recorded separately. PDFp1

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Performance fairness for heterogeneous inference models and goodput objectives are explicit. PDF pp1-2; sections1.1-1.4
Q2	2	2	Online solo-run goodput estimator, model-state arrays, temporal scheduling and feedback controller are specified. PDF pp5-8; section4; Algorithms1-3
Q3	1	1	Poisson load, sensitivity and trace duration are given; experiment repetitions and uncertainty remain unspecified. PDF pp8-9; sections5.1-5.4
Q4	2	1	DellR740,V10032GB,models,load andSLO are supplied; precise CPU/OS/driver/CUDA/runtime versions remain unreported. PDF pp2,8-9; Table1; section5
Q5	1	2	Clockwork,MPS+Clockwork and EDF conditions are explained; EDF intentionally lacks batching and is interpreted as a policy contrast. PDF p8; section5.1
Q6	2	2	Goodput is successful requests per second withinSLO; unfairness is max/min solo-to-shared goodput slowdown. PDF pp2,8-9; Eq1-2; Table3
Q7	1	2	SLO100ms,interval50ms,Poisson arrivals and saturation beyond1% deadline failures are specified; no numericalP99 latency is inferred. PDF p9; sections5.2-5.3
Q8	2	2	Two- and three-model local co-running experiments are distinguishable from61-model,fourGPU trace replay. PDF p9; sections5.3-5.4
Q9	1	2	Performance-isolation boundary is explicitly defined by unequal goodput slowdown and directly tested; no memory/security/fault guarantee is claimed. PDF p2; Eq1-2; PDF p9; Table3
Q10	1	1	Algorithms and memory-management reliance onClockwork are explicit; versioned reproduction setup/artifact detail remains incomplete. PDF pp5-8,10; section7

Outcome checks

E046 | SLO-compliant goodput: 524.29 requests/s. Value and context checked.

Baseline: Clockwork: 457.24 requests/s. Hardware: Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.. Workload: Local three-model load ratio 3:1:2.

Local outcome, not four-GPU replay or p99 Table3 row3:1:2 verifies524.29requests/s and unfairness1.05 versusClockwork457.24requests/s and1.50. Total600requests/s,SLO100ms,interval50ms. Not thefourGPU trace and not aP99 latency. Source anchor: PDFp9 / publishedp18;Table3;unfairness definitionPDFp2 / publishedp11;Eq2

E047 | Unfairness ratio: 1.05 ratio. Value and context checked.

Baseline: Clockwork: 1.50. Hardware: Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.. Workload: Local three-model load ratio 3:1:2.

Maximum/minimum model slowdown; not security isolation Table3 row3:1:2 verifies524.29requests/s and unfairness1.05 versusClockwork457.24requests/s and1.50. Total600requests/s,SLO100ms,interval50ms. Not thefourGPU trace and not aP99 latency. Source anchor: PDFp9 / publishedp18;Table3;unfairness definitionPDFp2 / publishedp11;Eq2

S017 / P019

ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

Access status: Full text inspected

Version checked: User-supplied publisher typeset PDF; title, DOI, year and journal matched on first page

DOI: 10.1007/s42514-023-00154-y

Source URL: https://doi.org/10.1007/s42514-023-00154-y

Original QA: 16/20. Proposed QA: 16/20.

Criterion	Review finding and source
IC1	Supported in accessed version. Peer-reviewed journal article, English, published2023. PDF p1
IC2	Supported in accessed version. API-intercepted resource limiting among GPU co-tenants. PDF pp10-11; section4.3
IC3	Supported in accessed version. Five inference models share resources with six training models. PDF pp4,11; Tables1-2; section5.1
IC4	Supported in accessed version. P99, QPS and goodput results inspectable with comparator. PDF pp11-15; section5
IC5	Supported in accessed version. Thirty local co-location pairs; four-GPU app-mix totals kept separate. PDF pp11-14; sections5.2-5.3
IC6	Supported in accessed version. 2023 is inside the submitted2020-2026 window. PDF p1

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Objective is QoS-preserving inference/training co-location. PDF pp1-2; section1
Q2	2	2	NVML monitoring, CUDA API interception, predictor and adjustment are specified. PDF pp6-11; sections4.1-4.3
Q3	1	1	Model pairs and measurements are described; repetitions and uncertainty remain unspecified. PDF pp11-16; section5
Q4	2	2	V100, host, CentOS, CUDA and cuDNN plus five inference and six training models are given. PDF pp4,11; Tables1-3
Q5	1	1	MPS/k8s comparators and workload pairs are relevant; exact MPS allocation quotas remain unresolved. PDF pp11-12; section5.2; Figures14-17
Q6	2	2	QPS, query yield and goodput are distinguished with explicit definitions. PDF pp12-14; sections5.2-5.3; equations2-3
Q7	2	2	Single-stream query size1, P99 and twice-solo QoS target are explicit. PDF pp4,11-12; sections3.4,5.1-5.2
Q8	2	2	Thirty inference/training pairs and adjustment ablation directly test interference. PDF pp11-16; Figures14-21
Q9	1	1	Performance restriction is tested, but stronger cross-context/error claims lack a direct fault/security experiment. PDF pp2-3,10-11,15-16; sections4.3,5.4
Q10	1	1	Prediction/limiter overhead and limitations are discussed; data are on request and artifact/software details remain incomplete. PDF pp15-16; sections5.5-5.6; data availability

Outcome checks

E048 | Pairs meeting twice-solo p99 QoS target: 30 tested pairs. Value and context checked.

Baseline: MPS Non-ArkGPU. Hardware: Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.. Workload: Five inference models paired with six training models.

Count of tested pairs, not a universal guarantee Text confirms all30 tested pairs meet twice-solo P99 target; not a universal guarantee. Four-GPU app-mix goodput is separate. Source anchor: PDF p11 / printed p314; section5.2; Figure14 on PDFp12

E049 | Low-contention QPS loss: about 5 percent. Value and context checked.

Baseline: Unthrottled co-location. Hardware: Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.. Workload: Low-contention tested pair settings.

Regression retained; larger system-level goodput gains are not one-GPU results Text reports about5% QPS loss in low-contention cases versus MPS; adverse outcome retained. Source anchor: PDF p12 / printed p315; section5.2; Figure16 on PDFp13

S018 / P020

Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

Access status: Full text inspected

Version checked: User-supplied publisher typeset PDF; title, DOI, year and journal matched on first page

DOI: 10.1007/s10586-025-05284-2

Source URL: https://doi.org/10.1007/s10586-025-05284-2

Original QA: 15/20. Proposed QA: 15/20.

Criterion	Review finding and source
IC1	Supported in accessed version. English peer-reviewed Cluster Computing article published2025. PDF p1
IC2	Supported in accessed version. MIG placement and dynamic GPU partitioning are specified. PDF pp8-11; section3.2
IC3	Supported in accessed version. Small-language-model inference jobs and batch ranges are identified. PDF pp11-12; Table2
IC4	Supported in accessed version. Local JCT/STP and prediction-error sensitivity are inspectable; not online P99. PDF pp11-14; sections4.1-4.2
IC5	Supported in accessed version. Sections4.2/4.3.1 use a single A30; section4.3.2 simulation kept separate. PDF pp13-14
IC6	Supported in accessed version. 2025 is inside the submitted2020-2026 window. PDF p1

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Fragmentation, placement and dynamic partitioning objective is explicit. PDF pp1-2; section1
Q2	2	2	Partition prediction, BnB scheduling, constraints and reconfiguration-cost model are described. PDF pp5-11; section3; Algorithms1-3
Q3	1	1	Workload counts, comparators and metrics are given; repeated-run uncertainty is not established. PDF pp11-16; section4
Q4	2	2	A30 system, driver, CUDA, DCGM, model variants and batch sizes are specified. PDF pp11-13; Table2
Q5	1	1	MISO is adapted and ORACLE has prior knowledge; comparisons require those qualifications. PDF p11; section4
Q6	2	2	STP is a normalized weighted partition-throughput sum; makespan and JCT are separate. PDF p11; section4
Q7	1	1	Job completion, deadlines and queues are described; online request-tail measurements are not supplied. PDF pp11-14; sections4.1-4.3
Q8	2	2	Single-A30 job/batch concurrency tests are distinguishable from the large-cluster simulation. PDF pp13-16; sections4.2,4.3.1; Figures11-13
Q9	1	1	MIG placement boundaries are defined but no direct adversarial/fault-containment test is supplied. PDF pp8-9,13-14; architectural constraints and local evaluation
Q10	1	1	Implementation and prediction/reconfiguration limits are disclosed; complete artifact/version details remain unresolved. PDF pp11,18; sections3.2.5,6.4

Outcome checks

E050 | Effect of 15 percent prediction perturbation: 8.09 percent makespan increase. Value and context checked.

Baseline: Unperturbed predictor. Hardware: Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon profiles2g.12gb+1g.6gb+1g.6gb.. Workload: Reported prediction sensitivity.

Sensitivity result; local JCT and cluster simulation are separate 15% prediction error gives8.09% longer makespan and two extra reconfigurations. This belongs to section4.2, not the section4.3.2 cluster simulation; job/makespan outcome, not request-tail latency. Source anchor: PDF pp13-14; section4.2; Figure12 on PDFp16

S019 / P021

KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

Access status: Full text inspected

Version checked: Author-hosted 14-page HPCA manuscript; IEEE DOI identity confirmed; exact publisher-byte equivalence unverified

DOI: 10.1109/HPCA56546.2023.10071121

Source URL: https://danielwong.org/files/KRISP-HPCA2023.pdf

Original QA: 16/20. Proposed QA: 18/20.

Criterion	Review finding and source
IC1	Supported in accessed version. Peer-reviewed HPCA2023 identity established; evidence version is author manuscript. PDF p1; title/authors; IEEE DOI metadata
IC2	Supported in accessed version. Kernel-wise spatial partitioning among inference workers. PDF pp5-9; sectionsIV-V
IC3	Supported in accessed version. Concurrent homogeneous and mixed-model inference workloads. PDF pp9-10; sectionVI-A; TableIII
IC4	Supported in accessed version. P95 latency and throughput are inspectable, subject to emulation calibration. PDF pp10-11; Figure13; TableIII
IC5	Supported in accessed version. One MI50 supports separately extractable co-tenant tests. PDF pp9-10; one MI50, multiple workers
IC6	Supported in accessed version. 2023 lies within the main publication window. IEEE HPCA2023 DOI metadata

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Kernel-wise spatial right-sizing for concurrent inference is explicit. PDF p1; abstract; sectionI
Q2	2	2	Kernel-scoped partition proposal and emulation through CU masks are described. PDF pp5-9; sectionsIV-V
Q3	1	1	Emulation correction is transparent; repeated-run uncertainty for headline results is not established. PDF pp9-11; sectionsV-B,VI
Q4	2	2	MI50, host, Ubuntu/kernel, ROCm and model/worker/batch settings are supplied. PDF pp9-10; sectionVI-A; TableIII
Q5	1	1	Model-wise policy permits CU overlap beyond prior isolation constraints; these cases are flagged and limit equivalence. PDF pp9-10; sectionVI-A
Q6	2	2	RPS normalization and energy-per-inference measurement are defined; adjusted results estimate native support. PDF pp9-11; sectionsV-B,VI-B
Q7	2	2	P95 latency and SLO evaluation are explicit under maximum-load inference. PDF pp10-11; TablesIII-IV; Figure13b
Q8	2	2	One, two and four workers plus mixed-model pairs exercise contention. PDF pp10-12; Figures13-15
Q9	1	2	KRISP-I prevents CU overlap and is compared with oversubscribed KRISP-O; this tests a compute-resource/performance boundary only. PDF pp10-12; KRISP-I/O definitions; Figures13,16
Q10	1	2	Emulation calibration, adjusted latency formula and proposed hardware changes permit informed appraisal of implementation limits. PDF pp7-9; sectionsIV-D,V-B

Outcome checks

E051 | Evaluated latency statistic: 95 percentile. Value and context checked.

Baseline: AMD unrestricted concurrency labelled MPSDefault. Hardware: One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5. Workload: Inference workers and heterogeneous pairs.

Statistic definition, not a 95-percent latency improvement 95 denotes P95, not effect size. MPS Default is an AMD unrestricted-concurrency analogue. Emulation correction preserves right-sizing/allocation overhead. Source anchor: Author PDF pp9-11; sectionVI; TableIII and Figure13b

S020 / P022

Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

Access status: Full text inspected

Version checked: arXiv v3, 28 February 2025; ASPLOS 2025 DOI/ISBN/conference footer present; publisher-byte equivalence not asserted

DOI: 10.1145/3669940.3707282

Source URL: https://arxiv.org/pdf/2410.07381v3

Original QA: 17/20. Proposed QA: 19/20.

Criterion	Review finding and source
IC1	Supported in accessed version. ASPLOS2025 identity matches the v3 manuscript title/authors/DOI; exact publisher-byte equivalence unverified. arXiv v3 PDF p1; ASPLOS2025 DOI/conference/ISBN
IC2	Supported in accessed version. API interception and thread-block priority scheduling share a GPU. arXiv v3 PDF pp5-8; section4
IC3	Supported in accessed version. High-priority inference plus training, and multi-inference tests, are separable. arXiv v3 PDF pp9-11; Table2; Figures5-6
IC4	Supported in accessed version. P99 and normalized total-system-throughput results are inspectable. arXiv v3 PDF pp9-10; sections5.1,5.3
IC5	Supported in accessed version. Same-GPU workload results are distinct from the eight-GPU host inventory. arXiv v3 PDF p11; shared-GPU co-location discussion
IC6	Supported in accessed version. 2025 lies within the main publication window. arXiv v3 PDF p1; ASPLOS2025

QA appraisal

Item	Old	Proposed	Rationale and anchor
Q1	2	2	Performance isolation for high-priority inference with best-effort co-tenants is explicit. arXiv v3 PDF pp1-2
Q2	2	2	Slicing, preemption, API mediation and profile-guided priority scheduling are described. arXiv v3 PDF pp5-8; section4
Q3	1	1	Profiler averages repeated measurements, but end-to-end repetitions and uncertainty remain unspecified. arXiv v3 PDF pp8-11; sections4.2,5.1-5.3
Q4	2	2	A100 SXM, host, runtime versions and six training/six inference models are explicit. arXiv v3 PDF p9; Table2; section5.1
Q5	1	2	Common workloads, adapted trace, load, hardware and normalized metrics are documented for the four comparison mechanisms. arXiv v3 PDF pp9-11; sections5.1-5.3
Q6	2	2	System throughput sums normalized throughput of all concurrent workloads; it is not background throughput alone. arXiv v3 PDF pp9-10; sections5.1,5.3
Q7	2	2	P99 latency, 50% end-to-end load and 10%-90% idle-time sensitivity are specified. arXiv v3 PDF pp9-11; Figures5-6
Q8	2	2	Inference/training pairs and scaling to multiple inference workloads are evaluated on a shared GPU. arXiv v3 PDF pp9-11; sections5.2-5.4
Q9	1	2	P99 and workload-pair tests directly assess performance isolation; no adversarial memory or cross-tenant fault guarantee is inferred. arXiv v3 PDF pp9-13; sections5-6
Q10	2	2	Artifact setup, virtualization/kernel-transformation overhead and CUDA cooperative-kernel limitations are documented. arXiv v3 PDF pp12-14; sections5.7,6; AppendixA

Outcome checks

E052 | Normalized total system throughput: 1.19 ratio. Value and context checked.

Baseline: TGS: 1.65. Hardware: Shared A100 SXM 40GB; eight-GPU EC2 p4d host inventory is not an eight-GPU outcome; Ubuntu20.04/CUDA12.2/PyTorch2.2.0/ONNX Runtime GPU1.17.0/Hidet0.3.0. Workload: Ten percent idle-time setting.

Sum of normalized throughput of all concurrent workloads: Tally1.19 versus TGS1.65 at10% idle time. Not background throughput alone. Values verified in v3; metric label corrected and version pinned. Source anchor: arXiv v3 PDF p10; section5.3; Figure6a on p11

Scope, novelty and recommendations

Prepared on 22 October 2026 Pakistan time by the AI assistant at Raazia's request for Rameen's assignment. This is a supplementary audit, not a rerun of the protocol's four formal database searches. Exact instrumented queries, time, source and returned references are in novelty_search_receipt.json. Search-result counts are not added to PRISMA identification totals.

Scope question for the instructor

Our Topic 23 first draft uses the supplied bounded corpus of 20 unique studies represented by 21 report versions. The inherited register records 4,189 occurrences, 114 duplicate links, 4,075 canonical records, 3,493 screening exclusions and 559 retained records that were not advanced for assessment. Twenty-three reports were source-assessed in the supplied baseline, with two report exclusions. The 559 records are unassessed; they are not technical exclusions or evidence that no further eligible studies exist.

Question: May we submit the first draft as an explicitly bounded review with this limitation while completing source verification, or must all 559 retained records undergo the remaining screening and full-text process before final submission? If bounded scope is acceptable, please confirm the permitted boundary and whether any particular mechanism, workload or date coverage must be added.

Instructor decision: Pending. Date, exact response and evidence location: Not supplied. No approval, waiver, exclusion decision or final corpus closure is implied by this prepared question. The question has not been sent.

Consequences of the decision

If bounded scope is accepted, preserve the dated response, state the coverage restriction in the abstract/methods/PRISMA/validity sections, and retain the ordered unassessed register. Continue to distinguish inherited assessments from fresh source verification. Acceptance of a first draft does not establish exhaustive final SLR coverage.

If full assessment is required, freeze the actual retained/unassessed IDs before allocating 187 to Eshaal, 186 to Raazia and 186 to Rameen. These are prospective workloads, not completed decisions. Apply the submitted criteria and real independent votes; obtain full text, QA and extraction for new eligible studies; recalculate the study/report mapping and PRISMA from actual records. Do not turn the 559 into automatic exclusions to keep the 20-study count.

Survey comparison

Yu et al., 2022, A Survey of Multi-Tenant Deep Learning Inference on GPU, arXiv:2203.09040. The eight-page full text was inspected. Its full-stack view includes service, graph, runtime, kernel and resource layers and discusses streams, MPS, MIG and hybrid scheduling. This is a directly overlapping contextual survey, not a primary study in our set. Source: PDF pp1-3 and sectionsIII-IV; https://arxiv.org/abs/2203.09040.

Huang et al., 2025, On Efficiency, Fairness and Security in AI Accelerator Resource Sharing: A Survey, ACM Computing Surveys57(9),Article221, DOI10.1145/3721427. The35-page author-hosted publication copy was inspected. It reviews2016-2024 work across AI accelerators and already organizes discussion around efficiency, fairness, interference and security. Those themes cannot independently establish novelty for our review. Its wider accelerator/workload remit differs from our single-physical-GPU inference boundary. Source: PDFpp1-4; https://www.cs.newpaltz.edu/~lik/publications/Jiahua-Huang-ACM-CSUR-2025 and https://doi.org/10.1145/3721427.

Ye et al., 2024, Deep Learning Workload Scheduling in GPU Datacenters: A Survey, DOI10.1145/3638757. Publisher, institutional and author metadata were identified. The final full text was not retrieved in this pass; the related2022preprint has a different title and author order, so it is not silently treated as the final article. Available abstract metadata covers training and inference scheduling and resource utilization. Author-hosted40-page text now confirms broad inference scheduling and hybrid co-location coverage (Figure1,Section4,Section5.2); placeholder citation metadata is not final publisher pagination. Source: https://doi.org/10.1145/3638757 and https://scholar.pku.edu.cn/yezhisheng/publications/deep-learning-workload-scheduling-gpu-datacenters-survey.

Mandrik, A comprehensive review of GPU virtualization and sharing methods, HALhal-05429292v1. Repository-index results date the technical report22December2025; aJanuary2026author-upload date also appears. Direct repository access was blocked. Treat it as a recent contextual lead with potentially strong mechanism overlap. Its exact full-text coverage and peer-review status are not verified. Source: https://hal.science/hal-05429292v1.

Contribution wording supported by this audit

The defensible contribution is a bounded, source-audited comparison of single-GPU inference-sharing evidence that keeps workload, metric, baseline, resource boundary and implementation cost attached to each recommendation, and incorporates relevant2025systems. This is a narrower analytical emphasis and update, not evidence that no prior review has asked similar questions. The novelty comparison remains qualified by the unavailable Mandrik technical-report full text and the unverified final publisher pagination of the Ye author copy and by supplementary-search coverage.

Proposed text for integration: "Prior surveys already cover multi-tenant GPU inference and accelerator-sharing efficiency, fairness and security. This review therefore focuses on the traceability and comparability of within-one-physical-GPU inference outcomes, distinguishing performance control from security and fault containment and linking engineering recommendations to their measured conditions. Its contribution is bounded by the supplied corpus and does not establish exhaustive novelty."

Recommendation and research-gap checks

REC02 has freshly inspectable XSched/Tally support for foreground latency versus total system throughput, subject to source conditions. REC04 includes ArkGPU and REC05 includes Dyna-P; their local findings were checked against the user-supplied publisher PDFs on22October2026. REC06 now includes the checked InferFair Table3 goodput524.29requests/s and unfairness1.05 under600requests/s,3:1:2mix,SLO100ms and50msperiod. These are conditional source outcomes, not a universal recommendation.

G01-G06 should remain observations about this reviewed corpus. Existing surveys discuss several of the same issues, and lack of a matched experiment in this set is not proof of absence throughout the literature. Dyna-P job completion and makespan are not substitutes for online request-tail measurements. KRISP's calibrated emulation is not a measured deployable NVIDIA MPS implementation. Tally'sE052quantity sums normalized throughput across concurrent workloads, rather than measuring background throughput alone.

Outstanding human review and access

All six assigned sources,60QAitems,36eligibility checks and9outcomes have been source-checked. InferFair was closed using the correct supplied publisher copy. Original forms remain frozen; the master now integrates the supported corrections.

Rameen's independent check of Eshaal'sP001-P007plusP023 awaits Eshaal's first-review return and actual separate human judgments. Rameen's own six-study work is to be independently checked by Raazia. AI preparation in this package is not either person's signature. Source-based scores are integrated in the current master; actual teammate adjudication remains pending.

Return files and remaining work

Rameen_Source_Review.xlsx contains nine tabs. CSVs preserve item-level identity, eligibility, QA, outcomes, corrections, survey overlap, recommendations and the independent-review queue. Review-specific scores for unavailable sources are blank; inherited values remain separately visible.

Next: record actual independent checks and disagreements; retain the documented559record scope limitation or complete the protocol screening queue. Rameen checks Eshaal’s P001-P007plusP023; Raazia checks Rameen’s six studies.

Review provenance: AI assistant preparing Rameen assignment at Raazia request. Checkpoint built: 2026-10-22T15:36:03.422680+05:00. Source hashes and retrieval receipts are retained alongside the records. Third-party full-text PDFs are excluded from public Git.